Problem 1. Express the running time of the following functions as summations.
YOU DO NOT NEED TO ANALYZE THE SUMMATIONS. JUST WRITE THEM DOWN.

(a)

```
function Func(n)
    x = 1
    for i = n to n^2 do
        for j = i to i^2 do
            for k = 1 to ⌊√j⌋ do
                x = x + ki − j
            end for
        end for
    end for
    return x
end function
```

$$\boxed{\sum_{i=n}^{n^2} \sum_{j=i}^{i^2} \sum_{k=1}^{\sqrt{j}} c}$$

(b)

```
function Func(n)
    x = 0
    i = n^3
    while i ≥ 1 do
        for j = 1 to ⌊log2(i)⌋ do
            x = x + i − j
        end for
        i = i − 8
    end while
    return x
end function
```

$\sum_{j=1}^{\log_2(i)} c$ $\quad (\log_2(i) \cancel{+1} - \cancel{1})c$

$\log_2(i) \cdot c$

# of iterations	value of i
0	n^3
1	$n^3 - 8$
2	$n^3 - 8 - 8$
⋮	⋮
k	$n^3 - 8k$

$i = 0$

$\frac{n^3}{8} - \cancel{8k} = 0 \quad \frac{+\cancel{8}k}{\cancel{8}} \qquad k = \frac{n^3}{8}$

$$c \sum_{k=0}^{n^3/8} \log_2(i)$$

$$\boxed{c \sum_{k=0}^{n^3/8} \log_2(n^3 - 8k)}$$

(c) Assume that $\texttt{Random}(i, j)$ returns a value between i and j uniformly randomly (including i and j).

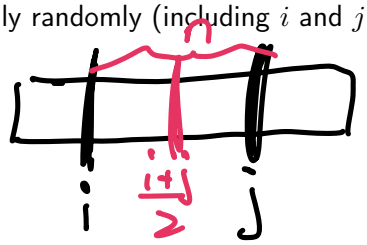
```
function Func(A,i,j)
    n = j - i + 1
    if n ≤ 1 then
        return 1
    end if
    for k = i to j do
        x = x + A[k]
    end for
    x = x + Func(A, i, ⌊(i+j)/2⌋)
    return x
end function
```

size of the array

c

$\sum_{k=i}^{j} c$ $\quad \underbrace{(j-i+1)}_{n} c$ $\quad cn$

$T\left(\frac{n+1}{2}\right)$ $\quad n = j - i + 1$



- $\left\lfloor \frac{i+j}{2} \right\rfloor - i + 1$
- $\frac{i}{2} + \frac{j}{2} - i + 1$
- $\frac{j}{2} - \frac{i}{2} + 1$

$(j - i + 2) \% 2$

$\boxed{\frac{n+1}{2}}$

$$\boxed{\begin{aligned} T(n) &= \cancel{c} + cn + T\left(\frac{n+1}{2}\right) \\ T(1) &= c \end{aligned}}$$

Problem 2. For the following algorithm,

(i) Express the best case running time as a recurrence relation Use $B(n)$ to denote the best case running time of the algorithm with an input of size n.

(ii) Express the worst case running time as a recurrence relation. Use $W(n)$ to denote the worst case running time of the algorithm with an input of size n.

YOU DO NOT NEED TO DETERMINE THE COMPLEXITY. SIMPLY WRITE DOWN A RECURRENCE RELATION AND THAT IS ALL.

```
function Func(A[],n)
    if n ≤ 10 then
        return A[1]
    else
        for i = 1 to n^2 do
            A[1] = A[1] + i − j
        end for
        k = Random(n)
        if k mod 8 = 5 then
            A[1] = A[1] + Func(n − 6)
        else
            A[1] = A[1] + Func(n/5)
        end if
        return A[1]
    end if
end function
```

THE BEST CASE IS ☆ NEVER THAT n IS SMALL.

$\sum_{i=1}^{n^2} c = cn^2$

a) $B(n) = cn^2 + B(n/5)$ $B(10) = c$

b) $W(n) = cn^2 + W(n-6)$ $W(10) = c$

Problem 3. Determine the asymptotic complexity of the following summation.

$$T(n) = \sum_{k=1}^{n^2} \sum_{i=0}^{k-1} c\sqrt{k-i}$$

UB:

$$T(n) \le \sum_{k=1}^{n^2} \sum_{i=0}^{k-1} c\sqrt{k}$$

$$T(n) \le \sum_{k=1}^{n^2} k\, c\sqrt{k}$$

$$T(n) \le \sum_{k=1}^{n^2} c k^{3/2}$$

$$T(n) \le \sum_{k=1}^{n^2} c (n^2)^{3/2}$$

$$T(n) \le (n^2) \cdot cn^3$$

$O(n^5)$

LB:

$$T(n) \ge \sum_{k=1}^{n^2} \sum_{i=0}^{k/2} c\sqrt{k-i}$$

$$T(n) \ge \sum_{k=1}^{n^2} \sum_{i=0}^{k/2} c\sqrt{k-\frac{k}{2}}$$

$$T(n) \ge \sum_{k=1}^{n^2} \left(\frac{k}{2}\right) c\sqrt{k/2}$$

$$T(n) \ge \sum_{k=1}^{n^2} c\left(\frac{k}{2}\right)^{3/2}$$

$$T(n) \ge \sum_{k=\frac{n^2}{2}}^{n^2} c\left(\frac{k}{2}\right)^{3/2}$$

$$T(n) \ge \sum_{k=\frac{n^2}{2}}^{n^2} \left(\frac{n^2/2}{2}\right)^{3/2}$$

$$T(n) \ge \underbrace{\left(n^2 - \frac{n^2}{2} + 1\right)}_{\left(\frac{n^2}{2}+1\right)} \left(\frac{n^2}{4}\right)^{3/2}$$

$\Omega(n^5)$

Thus, $T(n) \in \Theta(n^5)$.

Problem 4. Determine the asymptotic complexity of each of the following recurrence relations in Θ notation. Fully justify your solutions and express the complexity as $\Theta(n^k)$ or $\Theta(n^k \log(n))$ whenever possible.
If the algorithm is exponential, then simply find exponential lower bounds.
If you analyze the problem using a recursion tree, show at least two levels of the tree.
If you analyze the problem using substitution, show at least two substitutions.

(a)

$$T(n) = c\sqrt{n} + T(n/9) \qquad T(3) = c \tag{1}$$

$\Theta(\sqrt{n})$

Plug $\frac{n}{9}$ into ☆

$$T(n/9) = c\sqrt{\frac{n}{9}} + T\left(\frac{n/9}{9}\right)$$

Plug $n/9^2$ into ☆

$$T(n/9^2) = c\sqrt{\frac{n}{9^2}} + T\left(\frac{n/9^2}{9}\right)$$

$k=1$

$$T(n) = c\sqrt{n} + T(n/9)$$

$k=2$

$$T(n) = c\sqrt{n} + c\sqrt{\frac{n}{9}} + T(n/9^2)$$

$k=3$

$$T(n) = c\sqrt{n} + c\sqrt{\frac{n}{9}} + c\sqrt{\frac{n}{9^2}} + T(n/9^3)$$

In general,

$$T(n) = \sum_{i=0}^{k-1} c\sqrt{\frac{n}{9^i}} + T(n/9^k)$$

$$T(n) = \sum_{i=0}^{k-1} \frac{c\sqrt{n}}{3^i} + \cancel{T(3)}^{\,c}$$

$$T(n) = c\sqrt{n} \sum_{i=0}^{k-1} 1/3^i + c$$

$(1/3)^i$

Stops when

$$\frac{n}{9^k} = 3$$

$$n = 3 \cdot 9^k$$

$$\frac{n}{3} = 9^k$$

$$\log_9(n/3) = k$$

$$T(n) = c\sqrt{n} \sum_{i=0}^{k-1} (1/3)^i + c$$

$$T(n) \le c\sqrt{n} \sum_{i=0}^{\infty} (1/3)^i + c$$

$$T(n) \le c\sqrt{n} \underbrace{\left[\frac{1}{1-1/3}\right]}_{3/2} + c$$

$O(\sqrt{n})$

$$T(n) \ge c\sqrt{n}\left[\frac{1}{3}\right]^0$$

$$\ge c\sqrt{n}$$

$\Omega(\sqrt{n})$

$$T(n) \in \Theta(\sqrt{n})$$

≥ 1 chip + conq. $\Rightarrow$ exp. LB

(b)

$$T(n) = cn\log_4(n) + 2T(n-24) + T(n-2) \qquad T(12) = c \tag{2}$$

$T(n) \geq 2T(n-24) + T(n-2)$

$T(n) \geq 2T(n-24) + T(n-24)$

$k=1$ $\quad T(n) \geq 3T(n-24)$ ☆

$k=2$ $\quad T(n) \geq 3 \cdot 3T(n-48)$

$k=3$ $\quad T(n) \geq 3 \cdot 3 \cdot 3T(n-72)$

In general,

$T(n) \geq 3^k T(n-24k)$

$T(n) \geq 3^{\frac{n-12}{24}} \cancel{T(12)}^{c}$

$T(n) \geq 3^{\frac{n-12}{24}} \cdot c$

$\Omega(3^{n/24})$

exp

Plug $n-24$ into ☆

$T(n-24) \geq 3T(n-48)$

Plug $n-48$ into ☆

$T(n-48) \geq 3T(n-72)$

Stops when

$n-24k = 12$

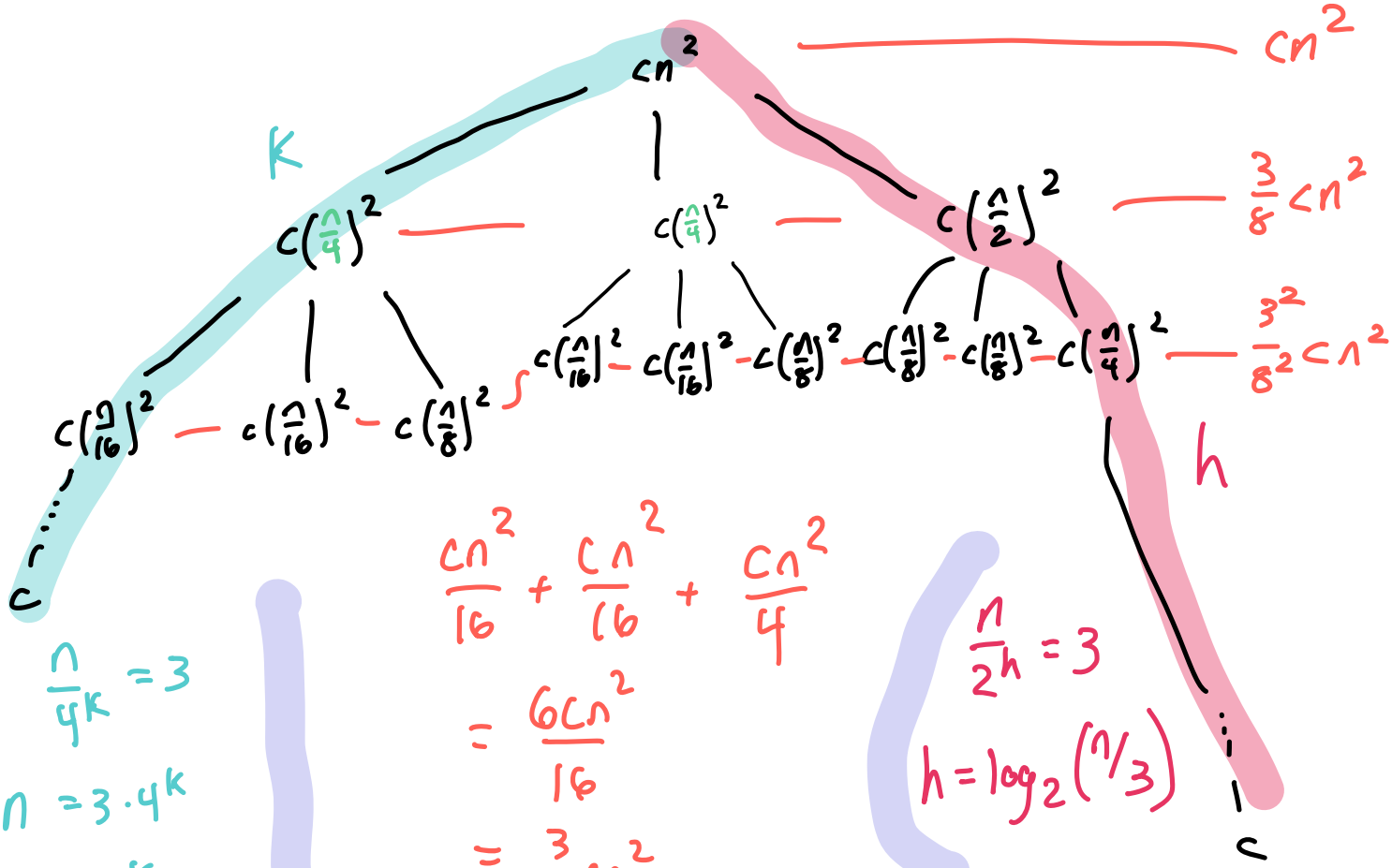
$n = 12+24k$

$n-12 = 24k$

$k = \frac{n-12}{24}$

(c)

$$T(n) = cn^2 + 2T(n/4) + T(n/2) \qquad T(3) = c \tag{3}$$



$$\frac{cn^2}{16} + \frac{cn^2}{16} + \frac{cn^2}{4}$$

$$= \frac{6cn^2}{16}$$

$$= \frac{3}{8}cn^2$$

$$\frac{n}{4^k} = 3$$

$$n = 3 \cdot 4^k$$

$$\frac{n}{3} = 4^k$$

$$\log_4(n/3) = k$$

$$\frac{n}{2^h} = 3$$

$$h = \log_2(n/3)$$

Cost @ i^{th} level $= \frac{3^i}{8^i} cn^2$

$$T(n) \geq \sum_{i=0}^{k} \frac{3^i}{8^i} cn^2$$

$$\geq cn^2 \sum_{i=0}^{k} (3/8)^i$$

$$\geq cn^2 \cdot 1$$

$$T(n) \leq \sum_{i=0}^{h} \frac{3^i}{8^i} cn^2$$

$$T(n) \leq cn^2 \sum_{i=0}^{\infty} (3/8)^i$$

$$\leq cn^2 \cdot \frac{1}{1 - 3/8}$$

$$\leq cn^2 \cdot 8/5$$

$$T(n) \in \Theta(n^2).$$

$$T(n) = cn^5 + 8T(n/2) + T(n/4)$$

$$\leq cn^5 + 8T(n/2) + T(n/2)$$

$$\geq cn^5 + 8T(n/4) + T(n/4)$$